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Multi-source trust model with uncertainty and context-temporal in cloud–edge–end systems

Minglong Cheng^{a,b}, Wei Chen^{a,b}, Chen Huang^a, Weidong Fang^c, Tingting Xu^{a,b,*},
Ziyang Xing^a, Zehua Wang^d

^a School of Computer Science and Technology/ School of Artificial Intelligence, China University of Mining and Technology, Xuzhou, 221116, China

^b Mine Digitization Engineering Research Center of the Ministry of Education, China University of Mining and Technology, Xuzhou, 221116, China

^c Science and Technology on Microsystem Laboratory, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai, 201800, China

^d Department of Electrical and Computer Engineering, The University of British Columbia, Vancouver, BC V6T 1Z4, Canada

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ABSTRACT

The openness of cloud–edge–end systems renders them vulnerable to malicious attacks, making accurate trustworthiness evaluation of edge nodes essential for reliable collaboration and quality-of-service assurance. Existing studies have overlooked the impact of trust evidence characteristics and cross-layer trust propagation on trust evaluation performance and attack detection capability. In this paper, a multi-source trust model based on uncertainty and context-temporal is proposed. Specifically, a multi-source trust evaluation mechanism is constructed from the cloud, edge, and end dimensions. In addition, multi-source trust is fused by considering epistemic uncertainty and conflict uncertainty to enhance evaluation robustness and reduce evaluation bias. Context similarity and a forgetting factor are introduced for weighted trust updating, enabling the model to preserve historical experience while improving adaptability to dynamic environments. Experimental results under various simulation scenarios demonstrate that the proposed model outperforms existing approaches in terms of accuracy, robustness, and response speed. In the task offloading scenario of cloud–edge–end systems, compared with the best-performing baseline, UCMT achieves average absolute reductions of 2.1, 2.4, and 2.3 percentage points in the offloading failure rate, SLA violation rate, and false positive rate, respectively.

1. Introduction

The exponential growth of IoT terminal devices and generated data has posed significant challenges to computation resources (Kumar and Agrawal, 2023). With the development of next-generation information technologies and artificial intelligence, low latency, high bandwidth, and abundant computation resources have become essential requirements for computation-intensive applications (Luo et al., 2026). The centralized computing paradigm of cloud computing suffers from inherent limitations in bandwidth and latency. In contrast, the distributed computing paradigm of edge computing deploys computation resources closer to end devices, thereby alleviating communication overhead and latency pressure (Apat et al., 2023). To achieve real-time and efficient intelligent computing and decision-making, cloud computing and edge computing have been integrated to form cloud–edge–end (CEE) computing systems (Yang et al., 2026).

CEE systems represent a dynamic distributed computing architecture that extends storage, communication, and computation resources

to the network edge to enable efficient collaborative computing (Wang et al., 2025b). However, due to resource constraints, heterogeneous network environments, and diverse user service requirements, appropriate computation offloading strategies must be developed to select suitable network resources for collaborative computing (Li et al., 2024). In this process, evaluating the reliability of edge servers (ESs) that provide computation resources is of critical importance, as they can access sensitive data from end users (EUs) (Liu et al., 2025). Furthermore, due to the openness of the network environment, malicious nodes may exist in the system. These nodes can launch attacks such as computation forgery, thereby affecting task processing efficiency and success rates and ultimately disrupting collaborative computing in the system (Wang et al., 2025d).

To address the security challenges in CEE systems, mechanisms such as encryption (Liu et al., 2026), authentication, and access control (Wang et al., 2025c; Chen et al., 2026) can mitigate issues including privacy leakage and external intrusion. However, they are unable to

* Corresponding author.

E-mail address: tingting.xu@cumt.edu.cn (T. Xu).

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defend against security threats caused by internal attacks (Dang et al., 2025). In internal attacks, malicious nodes have already bypassed these security mechanisms and obtained legitimate identities within the network. Therefore, a mechanism capable of identifying such malicious nodes is essential for reliable collaboration in CEE systems.

Trust models, which evaluate the reliability of nodes by analyzing their behavioral attributes and interaction records, have emerged as an effective security mechanism (Li et al., 2025). Their importance in CEE systems has been increasingly recognized (Song et al., 2025; W. Li et al., 2026). Sellami et al. (2025) and Xiang et al. (2025) employed blockchain-based approaches to enable traceable trust management, thereby enhancing the security of edge nodes and data. In addition, trust models have been leveraged to support collaborative training (Cheng et al., 2025) and computation offloading (Wang et al., 2025; Zhu and Wang, 2026) in edge computing environments. Trust models are widely used in CEE systems and provide security guarantees.

However, an analysis of existing trust models reveals several remaining challenges: (1) trust is typically evaluated through coarse-grained analysis and isolated computation at different layers, making it difficult to capture inter-layer trust propagation and intra-layer trust collaboration; (2) existing models lack the ability to adaptively respond to continuous behavioral changes of entities and exhibit limited resistance to coordinated attacks; and (3) the performance of nodes varies across different tasks, whereas existing models are unable to dynamically adjust trust judgments according to the current task context, nor can they effectively handle conflicts among trust evidence.

To address the above challenges, a multi-source trust evaluation model for CEE systems, named UCMT, is proposed. This model incorporates uncertainty-aware fusion and context-temporal adaptive update mechanisms to improve the accuracy and robustness. The practical decision supported by UCMT is the selection of ESs for computation offloading. By integrating trust evidence from multiple sources, UCMT provides an estimation of ES trustworthiness, enabling the CSC to preferentially allocate tasks to ESs with the highest overall reliability. As a core decision-making process in CEE collaboration, ES selection directly affects task completion quality and system efficiency. The main contributions are summarized as follows:

- A multi-source trust model is proposed to accurately model the trustworthiness of ESs. Each trust source provides evaluation bases in terms of long-term stability, collaboration reliability, and service quality. To mitigate potential malicious recommendations from users, their feedback behaviors are evaluated and incorporated as constraints in the trust computation.
- A dual-uncertainty trust fusion approach is proposed to enhance the robustness of multi-source evaluation. Trust contributions from different sources are adjusted according to epistemic and conflict uncertainties, reducing inaccuracies in trust evaluation.
- A context-temporal-aware trust update mechanism is developed. Context modulation ensures that evaluation results align with the current task environment, while a forgetting factor adjusts the weight of historical trust, providing a time-decayed representation of trust.

The remainder part of this paper is organized as follows. Related work is reviewed in Section 2. The system model and problem statement are presented in Section 3. The proposed UCMT model is detailed in Section 4. In Section 5, the experimental results are analyzed and discussed to validate the superiority of the UCMT model. Finally, the paper is concluded and future work is outlined in Section 6.

2. Related work

Nodes in CEE systems typically operate in open environments, making them highly susceptible to attacks. In addition, due to resource constraints, the presence of malicious nodes can significantly degrade

system efficiency (Zhu et al., 2024). Therefore, selecting reliable nodes to participate in collaborative tasks within CEE is an effective solution. In this section, existing research on trust models is reviewed from two categories: mathematical model-based and machine learning-based models.

2.1. Mathematical model-based trust model

Mathematical model-based trust models compute and update trust values using techniques such as probabilistic statistics and logical reasoning, and they are widely applied to real-time trust evaluation in CEE collaborative environments. Huang et al. (2026) updated node trust based on historical task success rates and collaborative feedback, while Mehjabin et al. (2024) assigned trust scores using physically unclonable functions to adapt to network dynamics. H. Yang et al. (2025) integrated the time dimension into edge node trust evaluation but failed to filter out malicious recommendations and relied on limited evaluation factors. Wang et al. (2025a) constructed a dual-layer trust framework from the user perspective, yet overlooked rich interaction records and dynamic updates. Liu et al. (2024) mitigated trust sparsity by computing service rating gaps over a sliding window, and enhanced recommendation rationality through multi-perspective weighting and reward-penalty mechanisms. Lin et al. (2025) evaluated entity trust comprehensively by considering collaboration degree, behavioral honesty, and content similarity. Langaroudi and Abdollahi Azgomi (2025) designed an anti-collusion trust model that evaluated entity trust values using QoS and QoSec. Yu et al. (2026) proposed a dynamic trust evaluation framework based on gray buffer states, which prevented abrupt trust drops under low-quality services. However, its state transitions relied on preset thresholds and lacked behavior adaptivity.

Such models are generally lightweight and highly interpretable. Nevertheless, existing approaches suffer from coarse-grained modeling of trust states, lack the capability to automatically adjust update rates or weight allocation based on overall historical interaction patterns, and exhibit limited resistance to coordinated attacks such as collusion.

2.2. Machine learning-based trust model

To address heterogeneity in multi-dimensional interactions of CEE systems, researchers have increasingly employed machine learning techniques to construct trust models. P. Li et al. (2026) achieved efficient resource-adaptive reasoning through meta-path-guided feature aggregation and attention mechanisms. Zhu et al. (2026) and Cheng et al. (2026) utilized graph neural networks to propagate and aggregate trust information across multi-hop neighbors, improving the accuracy of device reliability evaluation. However, such methods were prone to information explosion as network size increased. Sun et al. (2025) employed federated learning to develop a dynamic trust evaluation approach that continuously adjusted the selection of trustworthy entities. Farhat et al. (2025) assessed device trust by integrating communication, security, and higher-level attributes using artificial neural networks. F. Yang et al. (2025) monitored node behavior and communication outcomes through ensemble learning to predict trust values. Han et al. (2026) applied a reinforcement learning-based algorithm for trust evaluation, enabling adaptive weighting of trust evidence.

Machine learning-based models can automatically learn complex behavioral patterns and are highly adaptable to dynamic environments. Nevertheless, they depend on data distribution, lack interpretability, and struggle to meet the real-time requirements of resource-constrained nodes.

Table 1
Comparison of related work.

Models	Evidence source	Uncertainty	Trust update	Attacks	Trusted assumptions	Complexity
Huang et al. (2026)	End	●	●	●	Cloud	Low
Mehjabin et al. (2024)	End and Edge	○	○	○	Cloud	High
H. Yang et al. (2025)	End and Edge	●	●	●	Cloud and Edge (Re.)	Low
Wang et al. (2025a)	End and Edge	○	●	○	Cloud	Medium
Liu et al. (2024)	End	○	●	●	Cloud	Medium
Lin et al. (2025)	End	○	●	○	Cloud	Medium
Langaroudi and Abdollahi Azgomi (2025)	Edge and Cloud	●	●	●	Cloud	Low
Yu et al. (2026)	End	○	●	○	Cloud	Medium
P. Li et al. (2026)	End and Edge	●	●	●	Cloud and End	Low
Zhu et al. (2026)	End and Edge	○	○	○	Cloud and Agent	High
Cheng et al. (2026)	End and Edge	●	●	●	Cloud	High
Sun et al. (2025)	End	○	●	○	Cloud and End	Medium
Farhat et al. (2025)	End and Edge	○	●	○	Cloud and End	Medium
F. Yang et al. (2025)	Edge and Cloud	○	●	○	Cloud and Edge (Re.)	High
Han et al. (2026)	End and Edge	●	●	○	Cloud and End	Low
Our	End, Edge, and Cloud	●	●	●	Cloud	Low

Note: In uncertainty, ● represents addressing only conflict uncertainty, and ● represents addressing both conflict and epistemic uncertainty. In trust update, ● represents considering weight or time, ● represents considering events, and ● represents considering context and time. In attacks, ● represents considering collusion attacks, ● represents considering recommendation attacks, and ● represents considering all of the above. In trust assumptions, Re. denotes the recommendation node.

2.3. Summary

Given the diversity of CEE systems and their operating environments, evaluating the trustworthiness of ES entities is crucial for supporting collaborative operations within CEE. Existing trust models have contributed to computational efficiency and the capture of complex behavioral patterns. However, they still fail to meet the security requirements of CEE collaboration. Several issues remain to be addressed, including fine-grained trust analysis and trust propagation modeling, conflict handling among trust evidence, and the dynamic coupling between trust and task context. A detailed comparison of existing models is provided in Table 1.

To address above challenges, multi-source evidence from cloud, edge, and end layers is integrated to comprehensively evaluate ES trustworthiness. In addition, trust fusion and trust update mechanisms are established, enhancing the model's accuracy, robustness, and resilience to attacks in CEE systems.

3. Model and problem statement

3.1. System model

In this paper, the considered CEE system is illustrated in Fig. 1, comprising a Cloud Service Center (CSC), ESs, base stations, and EUs. The CSC serves as the central management layer of the system. Upon receiving service requests from EUs, it schedules the communication and computation resources of ESs and formulates appropriate offloading strategies. In addition, the CSC evaluates and maintains the trustworthiness of both ESs and EUs. The edge service layer consists of multiple ESs and base stations. ESs provide proximal computation resources to users, ensuring high-quality fulfillment of EU service requests. Base stations support ESs by facilitating communication between ESs and the CSC, while ESs can also communicate with each other to enable collaborative operations. The end-user layer represents the request sender, which offload locally intractable, computation-intensive tasks to ESs for processing. Each ES serves multiple EUs, and conversely, each EU can offload multiple tasks to multiple ESs.

In the CEE system, each entity locally maintains service records, including time parameters, resource parameters, and task parameters, while the CSC additionally records contextual parameters of tasks, such as task type, load status, and latency constraints. Due to the open nature of the system, malicious ESs and EUs may exist. Malicious ESs can launch attacks to forge computation resources and service records, thereby severely degrading service quality. Malicious EUs can launch recommendation attacks to intentionally inflate or deflate the actual trustworthiness of ESs.

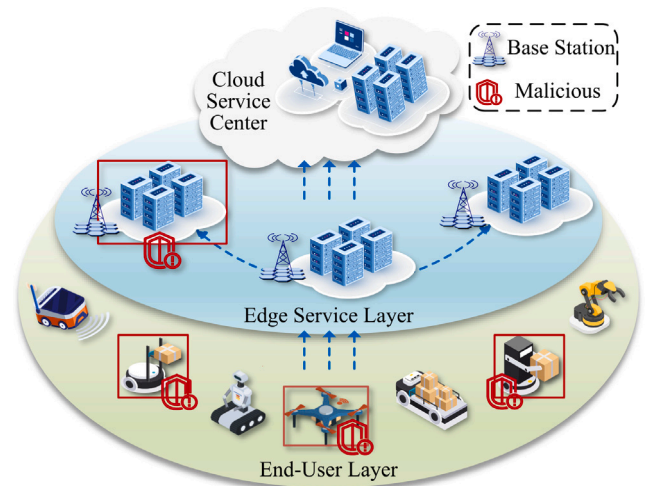


Fig. 1. System model.

The CSC is assumed to be secure and trustworthy. This assumption is not a simplification adopted for analytical convenience, but rather a fundamental prerequisite for establishing the proposed trust model. In trust evaluation, the absence of any trusted anchor would lead to a recursive dilemma in which an untrusted evaluator cannot produce trustworthy evaluation results. As the central management entity of the system, the CSC is responsible for policy coordination and global arbitration. Therefore, treating the CSC as a root of trust is both reasonable and necessary.

The UCMT model is designed to achieve accurate and robust trust evaluation against malicious behaviors originating from ES and EU. The effectiveness of UCMT in identifying such malicious nodes is based on the assumption that the CSC, acting as the trust root, faithfully executes the trust computation process and preserves the integrity of its observation records. The robustness claims of this work do not cover scenarios in which the CSC itself is compromised, since the CSC serves as the trust anchor in the proposed model. The potential consequences of CSC compromise are discussed separately in the limitations section (Section 5.9.3).

3.2. Problem statement

Let $E = (e_1, e_2, \dots, e_m)$ denote the set of ESs in the system, and let $U = (u_1, u_2, \dots, u_n)$ denote the set of EUs. The task request generated by

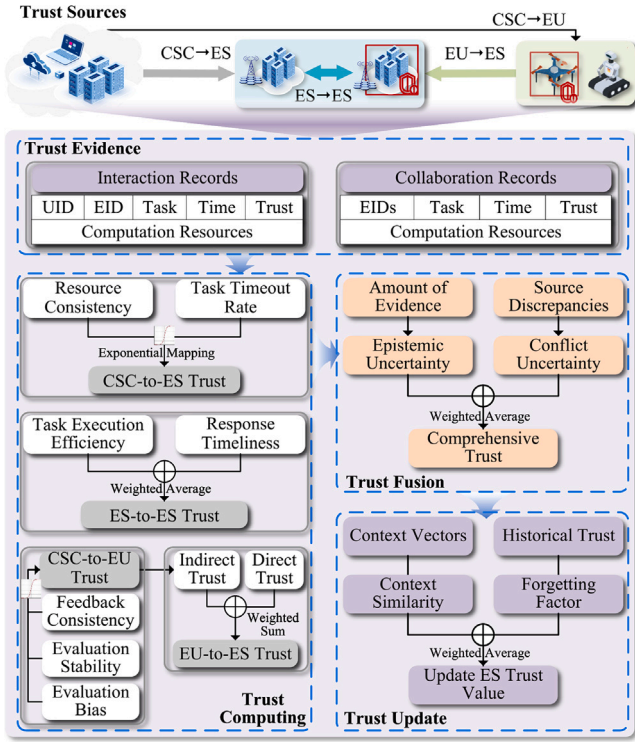


Fig. 2. UCMT model.

user u_n is represented as $X_n = (d_n, c_n, \tau_n)$, where d_n denotes the data size (in Mbits), c_n denotes the required computation capacity (in MFLOPs), and τ_n represents the maximum tolerable latency. Subsequently, the CSC communicates with ESs to obtain the currently available computation resources f_m (in MFLOPs/s). The core problem addressed in this paper is to construct a robust trust evaluation model in the presence of malicious behaviors. The proposed model is jointly determined by multi-source trust evidence and trust factors. Let $\mathcal{R} \in \{C, E, U\}$ denote the trust source set. The problem can then be formulated as

$$T_m^{com} = F \left(S_m^C, S_m^E, \left\{ S_{n,m}^U \right\}_{n \in \mathbb{N}_{U,E}}; \mu, Cr, \mathfrak{X}, \Delta t \right), \quad (1)$$

where $T_m^{com} \in [0, 1]$. $S_m^C, S_m^E, S_{n,m}^U \in [0, 1]$ represent the trust scores obtained from the CSC, collaborative ESs, and EUs, respectively. $\mathbb{N}_{U,E}$ denotes the set of EUs that have previously interacted with the target ES. μ represents epistemic uncertainty, which characterizes the insufficiency of evidence in trust evaluation. Cr denotes conflict uncertainty, measuring the degree of disagreement among different trust sources. $\mathfrak{X}$ represents task contextual information. Δt denotes the time interval used to characterize the temporal decay property of trust.

The mapping function $F(\cdot)$ represents a comprehensive trust mapping process jointly driven by uncertainty modulation and dynamic update mechanisms, with the primary objective of achieving robust estimation of ES trustworthiness.

4. UCMT model

In this section, a multi-source trust model, namely UCMT, is proposed to evaluate the reliability of ESs. The architecture of the UCMT model is illustrated in Fig. 2. The CSC, previously served EUs, and other collaboratively interacted ESs separately evaluate the trustworthiness of the target ES based on interaction and collaboration records. Then, the trust evaluations are fused according to trust certainty to form comprehensive trust, which is subsequently updated based on contextual information.

4.1. CSC-to-ES trust

As the global manager, the CSC possesses the capability to maintain long-term service records of ESs and focuses more on their overall reliability. It evaluates the trustworthiness of ESs based on resource status, task completion conditions, and service records. Whether the computation resources of an ES are sufficient is an important indicator for ensuring the completion of computation tasks. An apparently trustworthy node may adopt a selfish strategy of insufficient resource allocation for its own benefit. Although such behavior may not directly cause task timeout, it can significantly degrade service quality and impair user experience. Therefore, the resource sufficiency of an ES is first defined to measure whether the computation resources claimed by the ES are adequate to support task completion.

$$Rs_m = \frac{f_m^{alloc}}{f_n^{req} + \epsilon}, \quad (2)$$

where, ϵ is a small positive constant introduced to prevent the denominator from being zero, $f_n^{req} = c_n / \tau_n$ denotes the computation resources requested by the EU, and f_m^{alloc} denotes the computation resources allocated by the ES. c_n represents the computation capacity required by the EU, and τ_n denotes the maximum tolerable latency of the EU.

Furthermore, to characterize the long-term consistency of an ES in resource allocation behavior, the resource consistency Rc_m is derived by incorporating the resource sufficiency over multiple historical interactions.

$$Rc_m = \frac{1}{L} \cdot \sum_{i=1}^L \min \left(1, \frac{Rs_m^{(i)}}{Rs_{ideal}} \right), \quad (3)$$

where L denotes the number of historical records, and $Rs_{ideal} = 1.2$ represents the ideal resource sufficiency. This metric reflects whether the ES consistently maintains an honest resource allocation strategy.

In addition to selfish behavior, the ES may launch computation forgery attacks by falsely reporting idle resources, thereby causing task timeout or failure. Therefore, the task timeout rate ρ_m of an ES is further considered.

The task timeout rate of an ES is defined as the ratio of the number of timeout tasks $N_{task}^{timeout}$ to the total number of tasks N_{task}^{total} . This metric measures the execution outcomes of tasks and complements resource sufficiency, which focuses on the willingness of resource allocation.

$$\rho_m = \frac{N_{task}^{timeout}}{N_{task}^{total}}. \quad (4)$$

Since the CSC is trustworthy, the CSC-to-ES trust score is constructed solely based on objective indicators that can be directly observed by the CSC to ensure the effectiveness of the global trust reference. By jointly considering resource consistency and task timeout rate, the CSC-to-ES trust score S_m^C is obtained.

$$S_m^C = \frac{1}{1 + \exp(\kappa \cdot (\rho_m + (1 - Rc_m) - \theta))}. \quad (5)$$

where $\kappa = 10$ and $\theta = 0.55$ denote the scaling factor and shifting factor, respectively, which are used to control the nonlinear variation of the function. This design not only tolerates slight performance fluctuations but also rapidly decreases the trust score in response to severe violations.

The CSC-to-ES trust scoring model is designed as a nonlinear mapping mechanism, where a Logistic function is employed to transform accumulated risk factors into a normalized trust score. Specifically, multiple indicators are first converted into a unified risk representation to ensure consistent monotonicity. Subsequently, these risk components are aggregated, thereby introducing a nonlinear penalty.

4.2. ES-to-ES trust

In the CEE system, collaboration among edge nodes can improve overall system efficiency. Different ESs can request cooperation to complete the same task while mutually maintaining collaboration records. Therefore, when evaluating an ES, the trust provided by its collaboratively interacted ESs should also be considered. For clarity, the ES being evaluated is referred to as the collaborative ES, while the ES participating in the evaluation is referred to as the host ES.

ES-to-ES trust is defined as the evaluation of the collaborative ES by the host ES based on collaboration records, which include task response time, task completion time, requested computation capacity, and allocated computation resources. During the collaboration, response timeliness and execution efficiency serve as the primary evaluation factors.

First, the host ES e_k calculates the response timeliness $Rt_{k,m}$ of the collaborative ES e_m according to the task response time t_{res} . A more timely response from the collaborating ES indicates more reliable behavior.

$$Rt_{k,m} = \exp(-t_{res}), \quad (6)$$

Then, the task execution efficiency $Te_{k,m}$ of the collaborative ES is calculated. Based on the task completion time t_{fin} , the requested computation capacity c_e , and the allocated computation resources f_m^{ca} , the deviation between the actual task completion time and the expected completion time $t_{exp} = c_e/f_m^{ca}$ of the host ES can be determined. If the actual completion time does not exceed the expected completion time, the execution efficiency is regarded as optimal. Otherwise, the efficiency decays according to the relative timeout degree, thereby reflecting the trust degradation caused by the failure of the collaborative ES to complete the task on time.

$$Te_{k,m} = \exp\left(-\max\left(0, \frac{t_{fin} - t_{exp}}{t_{exp}}\right)\right). \quad (7)$$

In addition, the collaborative ES may provide different numbers of collaborations for different host ESs. Therefore, the collaboration familiarity $Cf_{k,m}$ between ESs is introduced as the aggregation weight for ES-to-ES trust. The collaboration familiarity is defined as the proportion of the number of collaborations $Num_{k,m}^{col}$ provided by the collaborative ES for a specific host ES to its total number of collaborations.

$$Cf_{k,m} = \frac{\log\left(Num_{k,m}^{col}\right)}{\sum_{i=1}^{|\mathbb{N}_c|} \log\left(Num_{i,m}^{col}\right)}, \quad (8)$$

where $\mathbb{N}_c$ denotes the collaboration set of the collaborative ES. A higher number of collaborations indicates that the evaluation result of the host ES is more reliable.

Accordingly, the comprehensive trust score of the host ES toward the collaborative ES is given as follows:

$$S_m^E = \frac{\sum_{i=1}^{|\mathbb{N}_c|} Cf_{i,m} \cdot \left(\frac{Te_{i,m} + Rt_{i,m}}{2}\right)}{\sum_{i=1}^{|\mathbb{N}_c|} Cf_{i,m}}. \quad (9)$$

4.3. EU-to-ES trust

The EU-to-ES trust is evaluated based on direct interaction records and recommendations from neighboring entities, consisting of two components: direct trust and indirect trust.

When an ES completes a task requested by an EU, the EU records the actual completion time t_n and its expected completion time $\hat{t}_n$ to calculate the direct trust score toward the ES.

$$S_{n,m}^{U,D} = \begin{cases} 0, & t_n > 2\hat{t}_n \\ 1 - \log\left(1 + \frac{t_n - \hat{t}_n}{\hat{t}_n}\right), & \hat{t}_n < t_n \leq 2\hat{t}_n \\ 1, & t_n \leq \hat{t}_n \end{cases}, \quad (10)$$

where $\hat{t}_n$ is calculated as follows:

$$\hat{t}_n = \frac{d_n}{r_{n,m}} + \frac{c_n}{f_m^{alloc}}, \quad (11)$$

where $r_{n,m}$ denotes the data transmission rate from the EU to the ES.

Eq. (10) is defined as a piecewise function, indicating that a larger deviation between the actual completion time and the expected completion time corresponds to a lower capability of the ES to complete the task. Although the case where the actual completion time exceeds the expected completion time is acceptable in CEE systems, a significantly longer actual completion time indicates that the ES is completely unreliable.

Furthermore, to improve the objectivity and comprehensiveness of EU-to-ES trust, indirect trust is additionally considered, namely, recommendations from other EUs toward the target ES. For users that have not previously interacted with the target ES, the introduction of indirect trust can also alleviate the cold-start problem. The indirect trust score is calculated as follows: where $\hat{t}_n$ is calculated as follows:

$$S_{n,m}^{U,R} = \frac{\sum_{k \in \mathbb{N}_{DR}, k \neq n} \omega_k \cdot S_{k,m}^{U,D}}{\sum_{k \in \mathbb{N}_{DR}, k \neq n} \omega_k}, \quad (12)$$

To suppress false recommendations, recommendation credibility is adopted to determine the recommendation weight ω_k , where the recommendation weight is equal to the recommendation credibility. The recommendation credibility is defined as the trust score of the user evaluated by the CSC based on records from the previous evaluation period, and its calculation is presented in Section 4.4.

The final EU-to-ES trust score is obtained through the continuous fusion of direct trust and indirect trust:

$$S_{n,m}^U = \omega_{n,m} \cdot S_{n,m}^{U,D} + (1 - \omega_{n,m}) \cdot S_{n,m}^{U,R}, \quad (13)$$

where $\omega_{n,m} = 1 - \exp(-N_{n,m}^D/N_{avg})$ denotes the trust fusion weight, which is determined by the number of direct interactions $N_{n,m}^D$ between the current EU and the target ES, as well as the average number of interactions N_{avg} between other EUs and the target ES. When the number of direct interactions is zero, the trust evaluation completely relies on recommendations. As the number of interactions increases, direct trust gradually becomes dominant.

4.4. CSC-to-EU trust

The CSC maintains the trustworthiness of each EU to measure the reliability of the feedback scores provided by the EU. This metric is dynamically updated based on the historical evaluation behaviors of the EU, including feedback consistency, evaluation stability, and evaluation bias.

The feedback consistency γ_n^{cons} is obtained by comparing the deviation between the task completion time reported by the EU and the corresponding task completion time recorded by the ES. It reflects the accuracy of the completion time reported by the EU.

$$\gamma_n^{cons} = \frac{1}{K} \cdot \sum_{i=1}^K \begin{cases} 1 - \log\left(1 + \frac{|t_i^n - t^E|}{t^E}\right) & 0 \leq t_i^n < 2t^E \\ 0 & t_i^n \geq 2t^E \end{cases}. \quad (14)$$

where K denotes the number of historical records in the EU.

The evaluation stability γ_n^{stab} measures the deviation between the recommendation values provided by the EU and the group consensus, thereby evaluating the similarity of recommendations. Recommendations that deviate from the group consensus are more likely to be malicious. Let $\mathbb{N}_{U,E}$ denote the set of EUs that have interacted with each target ES. The evaluation stability is calculated as follows:

$$\gamma_n^{stab} = 1 - \frac{1}{K} \cdot \sum_{i=1}^K \left| S_{n,E}^{U,D}(i) - \frac{1}{|\mathbb{N}_{U,E}|} \cdot \sum_{j=1}^{|\mathbb{N}_{U,E}|} S_{j,E}^{U,D}(i) \right|. \quad (15)$$

Relying solely on evaluating the feedback consistency and evaluation stability of EUs is insufficient to eliminate group collusion scenarios. Therefore, the evaluation bias β_n is further introduced, such that user feedback deviating from the CSC evaluation is explicitly penalized. This mechanism can suppress unreliable recommendations caused by both individual malicious users and malicious collusive groups.

$$\beta_n = \frac{1}{K} \cdot \sum_{i=1}^K |S_{n,E}^{U,D} - S_m^C|. \quad (16)$$

Finally, the CSC-to-EU trust score is defined as follows:

$$S_n^{U,ser} = \frac{1}{1 + \exp(5 \cdot ((1 - \gamma_n^{cons}) + (1 - \gamma_n^{stab}) + \beta_n - 1))}. \quad (17)$$

4.5. Comprehensive trust

Comprehensive trust is used to quantify the reliability of ESs in CEE systems. The CSC, ESs, and EUs are regarded as trust sources $\mathfrak{R} \in \{C, E, U\}$ of comprehensive trust.

During the calculation of comprehensive trust, different trust sources possess different amounts of evidence, resulting in different levels of understanding of the target ES. Consequently, uncertainty and conflict may exist among different evaluation results. For example, an evaluation result derived from ten interaction records is more reliable than that derived from only one piece of evidence. Furthermore, when the evaluation result of one evidence source is 0 while the evaluation results of all other evidence sources are 1, directly assigning the target trust value to 0 contradicts practical situations. To address this issue, a dual-uncertainty trust fusion method is constructed to decouple epistemic uncertainty and conflict uncertainty, thereby providing the basis for differentiated trust fusion strategies.

First, the epistemic uncertainty of trust sources is calculated. The epistemic uncertainty is defined as $\mu = \exp(-0.1 \cdot Num)$, where Num denotes the amount of evidence. For cloud trust and user trust, Num represents the cumulative number of tasks. For ES-to-ES trust, Num represents the number of collaborations. A smaller epistemic uncertainty indicates that the trust source is more reliable.

Conflict uncertainty is used to describe the degree of disagreement among different trust sources, which is defined as

$$C_{r_i} = \frac{1}{|\mathfrak{R}| - 1} \cdot \sum_{i,j \in \{C,E,U\}, i \neq j} |\mathfrak{R}(i) - \mathfrak{R}(j)|, \quad (18)$$

where, $\mathfrak{R}(U) = (\sum_{n \in \mathbb{N}_{U,E}} S_{n,m}^{U,D}) / |\mathbb{N}_{U,E}|$, $\mathfrak{R}(E) = S_m^E$, and $\mathfrak{R}(C) = S_m^C$. In Eq. (18), C_r represents the average degree of discrepancy between trust source i and the other two trust sources. During the calculation, i is fixed (e.g., when $i = C$, only the average discrepancy between CSC and the other two sources, ES and EU, is calculated). A lower conflict uncertainty implies that the corresponding trust source is more reliable. By incorporating conflict uncertainty, the fusion process can selectively reduce the weight of sources exhibiting significant deviations. Specifically, when two sources provide consistent evaluations while the third source substantially diverges, the fusion weight of the deviating source is suppressed, allowing the consensus information to retain greater influence.

Accordingly, based on CSC-to-ES trust, ES-to-ES trust, and EU-to-ES trust, the comprehensive trust score of the ES is obtained as follows:

$$S_m^{Com} = \frac{\sum_{i \in \{C,E,U\}} \mathfrak{R}(i) \cdot (1 - \mu_i) \cdot \exp(-3 \cdot C_{r_i}^2)}{\sum_{i \in \{C,E,U\}} ((1 - \mu_i) \cdot \exp(-3 \cdot C_{r_i}^2))}. \quad (19)$$

The fusion strategy in Eq. (19) is specifically designed for multi-source trust integration. The three trust sources considered in this work exhibit different characteristics and reliability levels, requiring a fusion mechanism capable of jointly handling heterogeneous evidence and uncertainty. Bayesian methods are commonly adopted in scenarios with binary and homogeneous interactions. In contrast, the trust evidence

in CEE systems consists of continuous and multi-dimensional behavioral indicators, including resource allocation, service performance, and collaborative behaviors. Directly reducing such information to binary outcomes may lead to the loss of behavioral details. Moreover, the trustworthiness of an ES is jointly determined by multiple trust sources, making cross-source consistency an important factor in trust evaluation. Therefore, UCMT adopts a deterministic multi-source fusion framework with uncertainty-aware modulation to accommodate the characteristics of heterogeneous trust evidence in CEE systems.

4.6. Trust update

The interaction records in a CEE system increase over time, and the behavior of entities changes continuously. Therefore, it is necessary to update the current trust value by incorporating historical trust records. However, the performance of an ES may varies under different task types and network conditions, and trust exhibits a nonlinear temporal decay property. To address these issues, a context modulation and forgetting factor update mechanism is introduced during the trust update stage, enabling the trust evaluation results to remain consistent with the current task environment.

The context vector is defined as $\mathfrak{X} = (dt_n, \xi_n, l_m)$, where the parameters are defined as follows:

- Based on the resource characteristics of a task, the task type is defined as $dt_n = c_n / (c_n + d_n)$. When $dt_n \rightarrow 1$, the task is computation-intensive, whereas when $dt_n \rightarrow 0$, the task is data-intensive.
- The latency constraint of the task is defined as $\xi_n = \tau_n / \bar{\tau}$, where $\bar{\tau}$ denotes the average task completion time of the ES. When $\xi_n \rightarrow 0$, the task has stringent latency requirements, whereas when $\xi_n \rightarrow 1$, the task has relaxed latency requirements.
- Based on the resource occupation of the edge server, the system load level is defined as $l_m = f_m^{alloc} / f_m^{total}$, where f_m^{total} denotes the total computation resources of the ES. When $l_m \rightarrow 0$, the ES is in an idle state, whereas when $l_m \rightarrow 1$, the ES is under a high-load condition.

For historical trust records, each record is associated with a context identifier $\mathfrak{X}_j$. Let the current context be $\mathfrak{X}^*$. The context similarity ϕ is calculated as

$$\phi(\mathfrak{X}_j, \mathfrak{X}^*) = \exp(-\|\mathfrak{X}_j - \mathfrak{X}^*\|_2). \quad (20)$$

Furthermore, to reflect the temporal decay characteristic of trust, a forgetting factor ψ is calculated using a forgetting curve.

$$\psi(\Delta t) = \frac{1}{1 + \alpha(\Delta t)^\beta}, \quad (21)$$

where Δt denotes the time interval between two updates. α and β are fitting parameters, which are set to 0.5 and 0.6, respectively. These parameter values can reasonably characterize the nonlinear temporal decay property of trust. In practical systems, they can also be adjusted according to the importance assigned to historical trust. Increasing α accelerates the decay rate, whereas increasing β causes faster decay in the early stage and slower decay in the later stage.

For any comprehensive trust value, its original trust value is the trust score obtained in Section 4.5. Based on historical context records, the comprehensive trust of the ES after the end of the J th period is updated as follows:

$$T_m^{com}(J) = \frac{\sum_{j=1}^J \psi(\Delta t_j) \cdot \phi(\mathfrak{X}_j, \mathfrak{X}_J) \cdot T_m^{com}(j)}{\sum_{j=1}^J \psi(\Delta t_j) \cdot \phi(\mathfrak{X}_j, \mathfrak{X}_J)}. \quad (22)$$

In summary, the execution process of UCMT is presented in Algorithm 1. The processes in which the CSC, host ES, and EU calculate ES trust correspond to Lines 2–3, Lines 4–6, and Lines 7–9, respectively. Lines 10–11 describe the trust fusion process, while Lines 13–16 correspond to the trust update process. The process by which the CSC calculates EU trust is presented in Lines 17–19.

Algorithm 1: UCMT Model.

Input : User set U , edge server set E , interaction and service records

Output: Comprehensive trust T_m^{com} for each $e_m \in E$

```

1 for each  $e_m \in E$  do
2   Calculate resource sufficiency  $Rs_m$ , resource consistency
    $Rc_m$ , and task timeout rate  $\rho_m$ 
3   Calculate CSC-to-ES trust score  $S_m^C$ 
4   for each host ES do
5     Calculate response timeliness  $Rt_{k,m}$ , Task execution
     efficiency  $Te_{k,m}$ , and collaboration familiarity  $Cf_{k,m}$  of
     collaborative ES
6     Calculate ES-to-ES trust score  $S_m^E$ 
7   for each  $u_n \in \mathbb{N}_{U,E}$  do
8     Calculate direct trust score  $S_{n,m}^{U,D}$  and indirect trust score
      $S_{n,m}^{U,R}$ 
9     Calculate EU-to-ES trust score  $S_{n,m}^U$ 
10  for each trust source  $\mathfrak{R} \in \{C, E, U\}$  do
11    Calculate epistemic uncertainty  $\mu$  and conflict
    uncertainty  $Cr$ 
12  Calculate comprehensive trust score  $S_m^{com}$ 
13  Retrieve historical trust records and task context records
14  for  $j = 1$  to  $J$  do
15    Calculate context similarity  $\phi$  forgetting factor  $\psi$ 
16  Update comprehensive trust value using Eq. (22)
17 for each  $u_n \in U$  do
18   Calculate feedback consistency  $\gamma_n^{cons}$ , evaluation stability
    $\gamma_n^{stab}$ , and evaluation bias  $\beta_n$ 
19   Calculate CSC-to-EU trust score  $S_n^{U,ser}$ 
20 return  $T_m^{com}$  for each  $e_m \in E$ 

```

4.7. Explainability analysis

The multi-source architecture of UCMT inherently supports interpretability. Although no dedicated interpretability module is explicitly designed, the structure of UCMT enables security analysts to trace the origin of a low comprehensive trust score through the following decomposition process.

For a low comprehensive trust score of an ES e_m , analysts can examine the contribution of each trust source T_m^{com} :

$$\Delta_i = T_m^{com} - \mathfrak{R}(i), \quad i \in \{C, E, U\}, \quad (23)$$

where Δ_i quantifies the deviation between the score of a trust source and the final trust score. A positive Δ_i indicates that the source score is lower than the final score and therefore acts as a major contributor to the low-trust evaluation. A negative Δ_i indicates that the source increases the final score, thereby partially offsetting the influence of other low-scoring sources.

Furthermore, the underlying factors within each trust source can be examined:

- CSC source: A low score indicates a high task timeout rate or poor resource consistency, suggesting that the ES consistently fails to complete tasks on time or allocates insufficient resources.
- ES collaboration source: A low score indicates poor response timeliness or low task execution efficiency, suggesting that the ES is unreliable in cross-edge collaboration.
- EU source: A low score reflects poor direct interaction records or low recommendation trust, indicating that the ES performs poorly in user-facing tasks or has received negative feedback from other users.

This decomposition provides actionable insights. If the CSC source is the primary contributor, administrators should investigate resource allocation policies or task scheduling strategies. If the EU source dominates, user experience or feedback reliability should be examined. If the ES source is the main contributor, greater attention should be paid to cross-edge communication or collaboration mechanisms.

5. Experimental analysis and discussion

In this section, the experimental settings are introduced, and the performance of UCMT is comparatively analyzed and discussed.

5.1. Experimental setup**5.1.1. Experimental parameters**

Simulation experiments are conducted on the Matlab R2024b platform. Unless otherwise specified, the parameters used in the experiments are listed in Table 2. In the system, the initial trust value of each entity is set to 0.5. During each evaluation period, a number of tasks with different types are randomly generated by EUs and assigned to ESs. GTOTC (Huang et al., 2026), LDTEM (H. Yang et al., 2025), FATE (Lin et al., 2025), CRTM (Langaroudi and Abdollahi Azgomi, 2025), and TrustGrey (Yu et al., 2026) are selected as baseline models for performance comparison with UCMT under the same environment.

For the statistical significance experiments, a two-level repeated design is adopted. At the outer level, 25 independent random seeds are used to construct 25 paired experimental units, and each experimental unit contains 100 Monte Carlo simulation instances. For each outer-level seed, the 100 internal results are first averaged to obtain a seed-level metric value. Under the same outer-level random seed, all methods use the same 100 sets of synthetic interaction evidence. Subsequently, the 95% confidence interval is calculated based on the 25 seed-level metric values (i.e., $1.96 \cdot std / \sqrt{25}$), and the Wilcoxon paired test is performed. The Wilcoxon test uses the difference between the seed-level means of two methods under the same outer-level random seed and the same synthetic evidence as the paired samples.

For the remaining experiments, 100 Monte Carlo repetitions are used to estimate the average performance under each experimental condition. Since these experiments are mainly used to demonstrate performance trends under variations in parameters, attack scenarios, offloading scales, or system overhead, no additional outer-level paired experimental units are constructed. Different methods still use the same synthetic interaction evidence under the same Monte Carlo index to ensure fair comparison.

The synthetic interaction evidence is generated under the control of fixed random seeds. For each experiment, edge servers, end users, malicious node identities, and server resources are generated first. Subsequently, task interaction evidence, user feedback evidence, and server cooperation evidence are generated according to the evaluation cycles. The execution, reporting, and feedback of normal nodes fluctuate slightly around their true values, while malicious nodes trigger resource misreporting, execution slowdown, or completion-time forgery according to their probabilities of malicious behaviors.

Under the same experimental settings, attack parameters and random seeds remain fixed across all competing methods. The evidence generated under the same random seed is simultaneously provided as input to UCMT and all baseline models, ensuring that the differences among different models are solely attributed to their trust evaluation mechanisms.

Table 2

Parameter settings.

Parameters	Value
Trust evaluation periods	100
Monte Carlo iterations	100
Number of ESs	50
Number of EUs	100
Percentage of malicious entities	20%
Data size	2–10 Mbits
Number of tasks per EU per period	2–6
Tolerance delay	2–6 s
Transmission rate	3–12
Computation capacity	2–10 MFLOPs
Computation resources	8–16 MFLOPs/s

5.1.2. Evaluation metrics

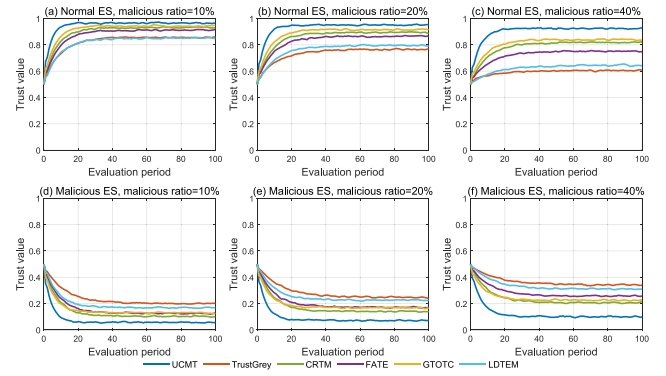
In Section 5.7, four metrics are employed, namely detection latency, CPU time, CPU utilization, and peak memory. Detection latency refers to the average wall-clock time required by the model to complete one trust evaluation for a single node within a single cycle, which is used to measure the response speed of online detection. CPU time represents the processor time consumed by the model to complete a complete multi-cycle trust evaluation, reflecting the actual computational cost of the algorithm. CPU utilization indicates the average proportion of the overall CPU resources occupied by the model during execution. Peak memory refers to the maximum memory consumption of the core state variables maintained during the trust evaluation process, which is used to measure the spatial overhead.

In Section 5.8, six metrics are adopted, including the offloading failure rate, SLA violation rate, malicious ES selection ratio, system utility, false positive rate, and false negative rate. The offloading failure rate represents the proportion of task failures caused by factors such as ES execution crashes or execution refusal. The SLA violation rate denotes the proportion of tasks that either fail or exceed the maximum tolerable end-to-end completion latency. The malicious ES selection ratio represents the proportion of offloading decisions selecting malicious ES among all offloading selections. System utility is a normalized comprehensive evaluation metric, which is calculated as the weighted sum of the normalized optimization objective score, offloading failure rate, SLA satisfaction, secure offloading capability, and normal ES utilization rate. The false positive rate is defined as the proportion of benign ES nodes that are incorrectly identified as malicious. The false negative rate is defined as the proportion of malicious ES nodes that are incorrectly identified as benign.

Furthermore, in the abstract, the improvement achieved by UCMT in terms of the offloading failure rate, SLA violation rate, and false positive rate is quantified using the average absolute reduction. The average absolute reduction is defined as the arithmetic mean of the absolute percentage point differences between UCMT and the best baseline in each scenario for each metric.

5.1.3. Attack settings

To evaluate the attack detection capability of the proposed model, six attack types, namely on-off attacks, collusion attacks, computation forgery attacks, malicious recommendation attacks, gradual infiltration attacks, and hybrid attacks, are introduced into the interaction process. Sudden attacks are modeled through behavior transitions. These attacks manifest in different forms of evidence, including resource allocation information, task completion time, reporting time, and trust scores. Malicious ES nodes may falsely advertise resource availability, intentionally delay task execution, and report optimistic completion times to the CSC with a predefined probability. Malicious EU nodes manipulate the trust scores of target ES nodes by injecting deceptive evidence into the recommendation chain. The probabilities of malicious behaviors for malicious ES and malicious EU are set to 0.82 and 0.78, respectively. The resource misreporting factor is 1.25, the execution slowdown factor is 1.8, and the malicious recommendation bias strength is 0.45.

**Fig. 3.** Comparison of trust convergence.

The on-off attack is modeled by assigning intermittent malicious behaviors to ES nodes. Specifically, malicious behaviors are performed with a high probability during the first three cycles, followed by reduced malicious intensity in the subsequent three cycles. For the collusion attack, the probability of malicious execution by ES nodes and the feedback bias probability of malicious EU nodes are simultaneously increased, such that service degradation and recommendation contamination occur concurrently within the same cycle. The computation forgery attack mainly targets resource misreporting and execution delays, thereby exposing distortions in resource fulfillment and timeout evidence used by the CSC for trust evaluation. The malicious recommendation attack deliberately manipulates the trust scores of target ES nodes by increasing or decreasing their reported trust values. The gradual infiltration attack follows a nonlinear escalation pattern, including a latent stage, an accelerated infiltration stage, and a short camouflage window. In the hybrid attack scenario, persistent low-quality service behaviors, computation/time forgery, and malicious recommendation collusion are jointly present.

5.2. Comprehensive performance

The convergence of the trust evaluation results for each model over the evaluation period, under different proportions of malicious entities, is shown in Fig. 3. In Fig. 3(a–c), the evaluation targets are normal ESs, whereas in Fig. 3(d–f), the evaluation targets are malicious ESs.

The convergence results under different malicious entity proportions demonstrate that UCMT maintains reasonable trust evaluation trends for both normal ES and malicious ES. As the evaluation periods increase, trust evidence gradually becomes richer. By adopting a multi-source and multi-factor trust evidence collection method, UCMT converges more rapidly than the baseline models and reaches a steady-state trust value that is closer to the actual behaviors of the ES within fewer evaluation periods. Furthermore, the comprehensive trust in UCMT is obtained through differentiated fusion driven by evidence-quantity-based epistemic uncertainty and cross-source-disagreement-based conflict uncertainty. Therefore, as the malicious ratio increases, UCMT preferentially suppresses evidence sources with high conflict and insufficient evidence, thereby avoiding interference from forged behaviors or false recommendations in ES trust evaluation.

In Figs. 4 and 5, the mean trust values and standard deviations of all normal ESs and malicious ESs are statistically analyzed. TrustGrey is sensitive to heterogeneous interactions under large-scale environments, resulting in relatively poor stability. The accumulation of observation noise and prediction bias in FATE more easily amplifies fluctuations. The stability of CRTM highly depends on the recommendation verification chain, and its standard deviation convergence remains limited when multi-source coupling is insufficient. GTOTC improves the filtering capability for unstable nodes through behavior partitioning

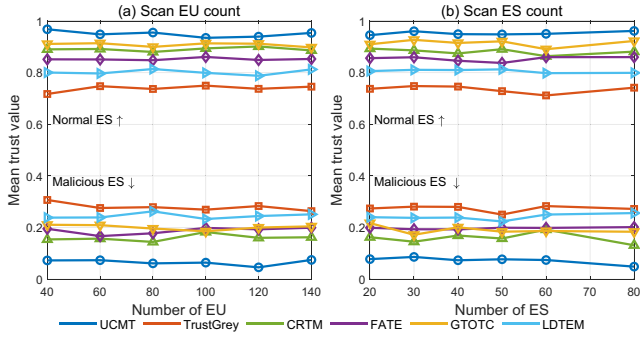


Fig. 4. Comparison of comprehensive trust mean value.

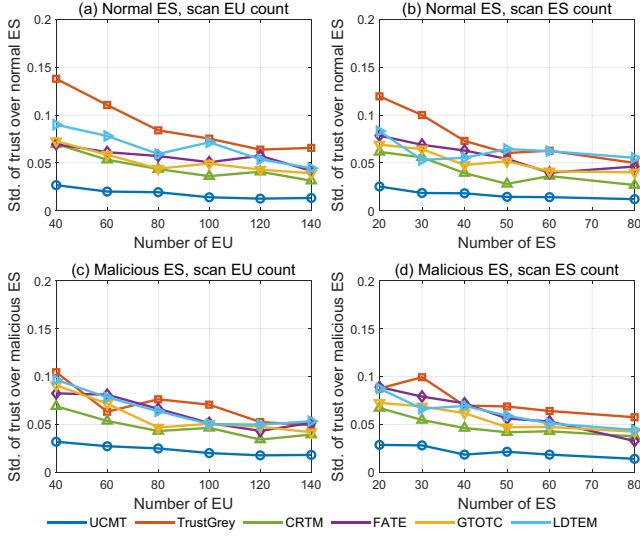


Fig. 5. Comparison of comprehensive trust standard deviation.

and temporal decay modeling, but still suffers from insufficient cross-layer evidence fusion. The Newton decay mechanism in LDTEM causes the trust values of normal nodes with insufficient long-term interactions to decrease. This effect becomes more evident when the scale of EUs/ESs increases, as the reduced number of observations for individual nodes makes LDTEM more likely to interpret insufficient evidence as moderate trust.

In contrast, as the scales of EUs and ESs vary, UCMT exhibits superior performance in both trust discrimination capability and fluctuation control capability. This is reflected in the more stable separation between normal and malicious ESs and the smaller dispersion during cross-scale migration. The primary reason is that recommendation information is not directly incorporated into the fusion layer but is first constrained by the trustworthiness of recommendation entities. Meanwhile, historical evidence is reinforced only under context-consistent conditions. Consequently, the heterogeneity and recommendation bias introduced by scale expansion do not significantly affect comprehensive trust. The experimental results demonstrate that UCMT achieves better scalability and stability than the baseline models.

To verify the statistical robustness of the comprehensive performance comparison results, further statistical significance analysis based on independent random seeds is conducted. The results in Fig. 6 show that UCMT maintains high trust values for normal nodes while assigning the lowest trust values to malicious nodes. Both types of evaluation results are statistically significant. UCMT achieves a balance between stability and usability, and the confidence intervals demonstrate the reliability of the experimental results.

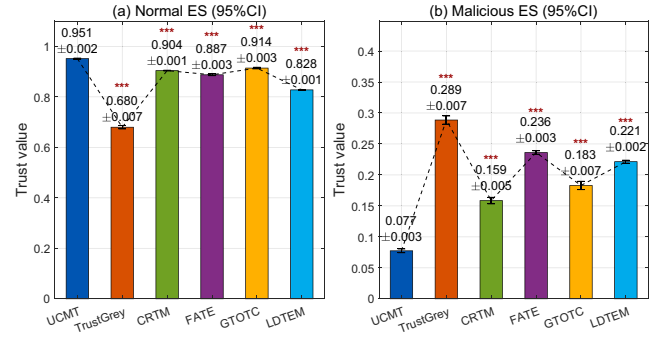
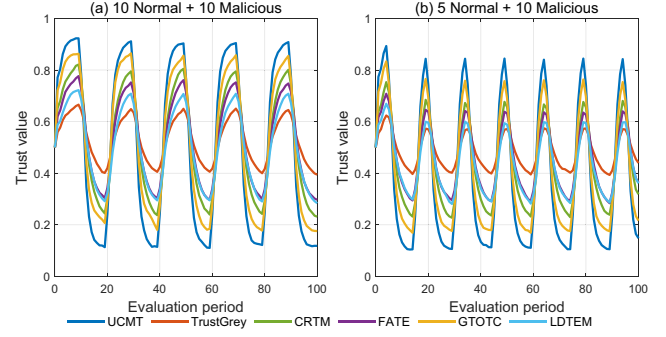
Fig. 6. Statistical tests. (vs. UCMT: * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$).

Fig. 7. Continuous behavior pattern transition.

5.3. Robustness evaluation

To evaluate the robustness and response speed of the proposed model, the trust variations of different models under behavior transition scenarios are compared. The continuous behavior transition patterns are illustrated in Fig. 7, where a single ES periodically alternates between 10 normal and 10 malicious periods, as well as 5 normal and 10 malicious periods. The one-time behavior transition scenario is illustrated in Fig. 8, where the behavioral attributes of the ES switch at the 15th period.

The periodic behavior transition scenarios impose stricter requirements on the dynamic tracking capability and robustness of trust models. Under such settings, UCMT exhibits the most pronounced rapid-rise and rapid-fall response characteristics. In UCMT, historical evidence is not retained with fixed weights over time but decays according to context similarity and time intervals, ensuring that recent behaviors dominate trust updates. GTOTC is affected by historical trailing effects, resulting in a slower response compared with UCMT. The time adjustment mechanism in CRTM limits its ability to rapidly respond to behavioral changes. FATE suffers from switching delays due to the persistence of predictive components. TrustGrey exhibits stronger smoothing effects under frequent switching, leading to slower adaptation to abrupt behavior changes. Overall, the results in Fig. 7 demonstrate that UCMT is more sensitive to behavioral dynamics and more robust under rapidly changing scenarios.

As shown in Fig. 8, when the ES transitions from malicious to normal or from normal to malicious, UCMT consistently adapts within a small number of evaluation periods, aligning the trust value with the current behavior state. When the current contextual observation significantly deviates from historical contexts, UCMT reduces the weight of historical evidence, allowing newly observed behavior to dominate the update process. These results further validate the effectiveness of UCMT in balancing historical memory and real-time evidence.

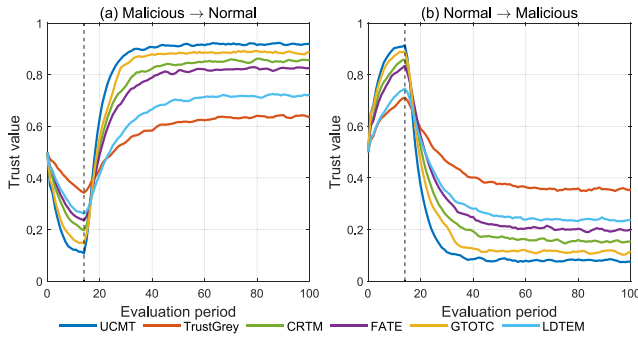


Fig. 8. One-time behavior pattern transition.

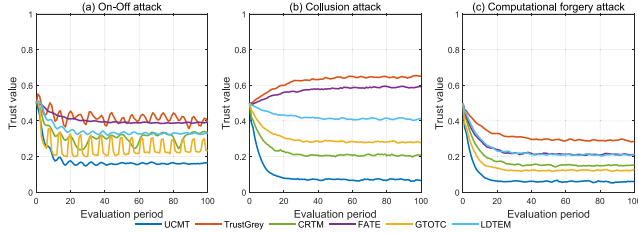


Fig. 9. on-off attack, collusion attack, and computation forgery attack scenarios.

5.4. Performance under various attacks

Under on-off attacks, collusion attacks, computation forgery attacks, malicious recommendation attacks, gradual infiltration attacks, and hybrid attacks, the detection capability of UCMT is evaluated.

The trust evolution under on-off attacks is illustrated in Fig. 9(a). TrustGrey exhibits rapid trust recovery due to its gray-state restoration mechanism, while FATE retains intermediate trust levels due to prediction inertia. Both methods fail to consistently maintain the target ES in a low-trust state over time. GTOTC demonstrates better performance than the above methods by capturing oscillatory behavior through sequential classification. In this scenario, the ES periodically launches malicious behaviors while its identity remains unchanged. As a result, even during inactive phases, the ES is still regarded as malicious by UCMT, leading to a smoother convergence toward low trust values.

Fig. 9(b) and (c) present the results under collusion attacks and computation forgery attacks, respectively. Both TrustGrey and FATE lack effective mechanisms for filtering coordinated fraudulent evaluations, resulting in inflated trust values. CRTM achieves better detection performance than most baselines by leveraging QoRec and honesty-aware mechanisms to mitigate collusion effects. In the case of computation forgery attacks, all baseline models are able to identify abnormal behavior to some extent. However, their steady-state trust values remain significantly higher than that of UCMT.

The trust evaluation results over evaluation periods under gradual infiltration attacks and hybrid attacks are shown in Fig. 10. In the gradual infiltration attack scenario, the behaviors of malicious ESs remain close to normal in the early stage, and the trust scores given by all models increase to some extent. As resource insufficiency, execution delays, and reporting inconsistencies gradually intensify, UCMT starts to rapidly decrease the trust of malicious ESs from the middle stage and remains lower than all baseline models during the subsequent convergence process. The buffering mechanism of TrustGrey and LDTEM makes their evaluation more conservative, the filtering prediction of FATE exhibits apparent inertia, CRTM has certain capability in handling recommendation quality and collusion behaviors but is less sensitive to resource fulfillment anomalies and multi-source conflicts, and GTOTC can capture long-term degradation trends but

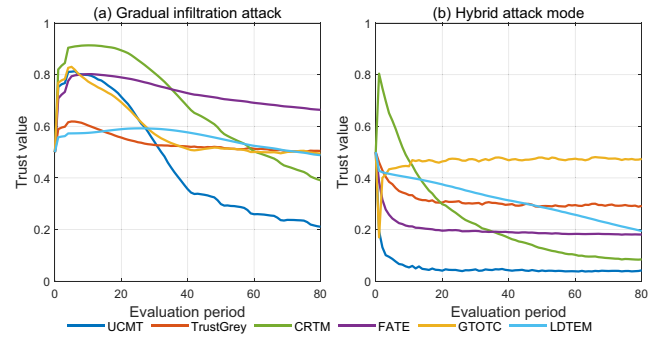


Fig. 10. Gradual infiltration attack and hybrid attack scenarios.

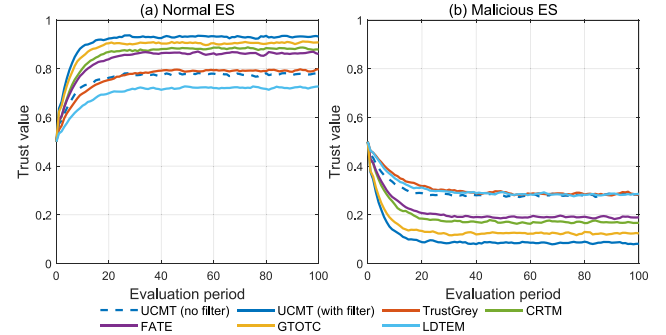


Fig. 11. Recommendation attack scenario.

relies on relatively coarse-grained three-state sequences. In the hybrid attack scenario, the conflict uncertainty mechanism of UCMT suppresses recommendations that deviate from mainstream evidence, while the time-aware update mechanism ensures that recent attack behaviors receive higher weights. Therefore, the trust of malicious ESs decreases rapidly and remains stable under UCMT. CRTM can identify some collusion behaviors but is still limited by the aggregation of direct and indirect trust. FATE, LDTEM and TrustGrey decrease more slowly due to their smoothing or buffering mechanisms. GTOTC responds to continuous failures to some extent but lacks fine-grained modeling of recommendation contamination and multi-source conflicts.

Overall, the results in Figs. 9 and 10 demonstrate that UCMT achieves higher consistency and accuracy under complex attack conditions. By jointly constraining resource consistency, task timeout rate, and cooperative execution efficiency, UCMT is able to detect forged behaviors at an earlier stage. In particular, under multi-source collaborative interference scenarios, UCMT effectively suppresses trust deviations caused by abnormal feedback.

The trust convergence under malicious recommendation attacks is illustrated in Fig. 11. The ablation-style comparison further verifies the effectiveness of using recommendation credibility to determine recommendation weights. Before contributing to comprehensive trust, recommendation information is constrained by consistency, stability, and bias evaluation, thereby reducing the contamination caused by false recommendations. In contrast, baseline models with weaker constraints are more prone to trust distortion. Among the baseline models, TrustGrey exhibits the largest trust deviation. This is because it lacks any mechanism for evaluating the credibility of recommenders and therefore cannot filter malicious recommendations.

5.5. Sensitivity analysis

In this section, the sensitivity of UCMT with respect to model parameters and task environments is analyzed, including the control

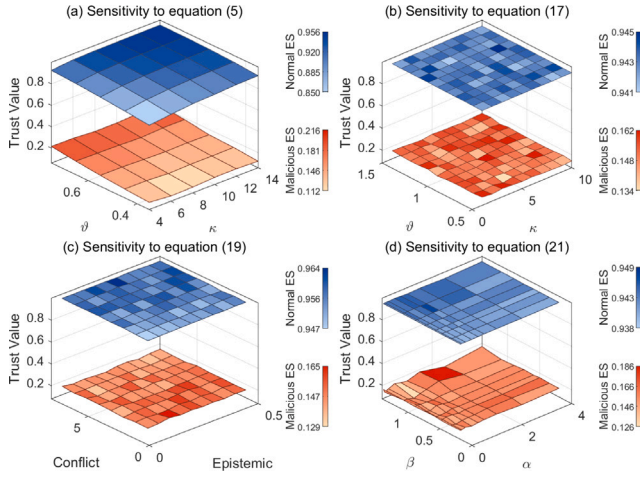


Fig. 12. Parameter sensitivity analysis.

parameters designed in key equations, penalty terms, ideal resource sufficiency, sparse interactions, cold start, and non-IID tasks.

To investigate the dependence of UCMT on the coefficients in its equations, a two-dimensional joint parameter sweep is conducted for the parameters in Eqs. (5), (17), (19), and (21). As shown in Fig. 12, the trust surfaces of normal ESs and malicious ESs maintain relatively small fluctuations, while the two types of nodes remain stably separated. The results indicate that the evaluation capability of UCMT does not heavily depend on the precise settings of these parameters and exhibits strong robustness to the coefficients in its equations. The control parameters mainly adjust the trend of nonlinear variations. They serve the intrinsic properties of trust rather than changing with different system categories.

Fig. 13 presents the sensitivity test of UCMT under different task environments. The robustness of UCMT is consistent with its architecture. For non-IID task distributions, all multi-source trust components are normalized ratios according to task requirements, and context modulation matches historical records based on task types, allowing heterogeneous tasks to be naturally accommodated. Therefore, the trust evaluation results remain almost unchanged. For ideal resource sufficiency, resource sufficiency only guarantees computation resource redundancy and complements the task timeout rate, allowing the model to remain robust. Under sparse interactions, normal ESs and malicious ESs gradually move closer to each other, leading to reduced discrimination. Cold-start nodes need to accumulate evidence over multiple evaluation periods, resulting in gradual convergence with slight residual errors. These results show that UCMT is robust to perturbations covered by normalized components and context modulation, while perturbations directly affecting evidence sufficiency exhibit directionally correct and predictable sensitivity. Nevertheless, such sensitivity remains within an acceptable range from the perspective of trust evaluation results.

The above results indicate that, within the simulation conditions and parameter variation ranges considered in this study, UCMT maintains robustness in terms of evaluation performance with respect to model control parameters and diverse task environment variations. Nevertheless, this robustness conclusion is derived from a specific simulation setting. In practical cross-scenario and cross-architecture deployments of CEE systems, the generalization capability of the model parameters may be affected by factors such as physical topology, data distribution, and node heterogeneity. Therefore, further adaptation and validation are required according to specific deployment environments.

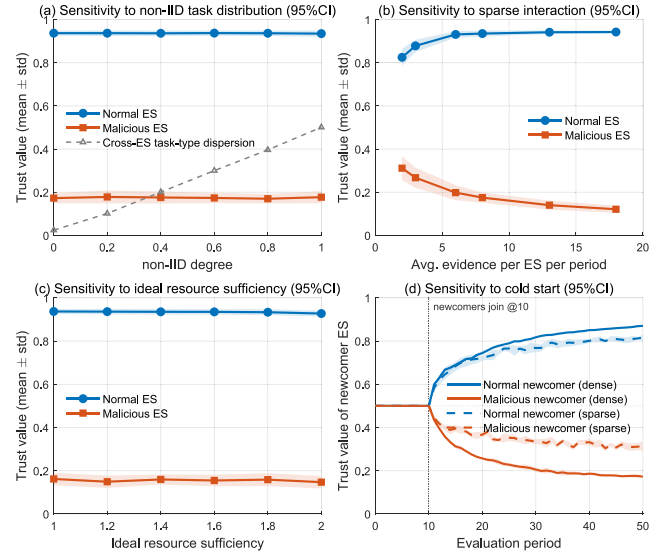


Fig. 13. Task environment sensitivity analysis.

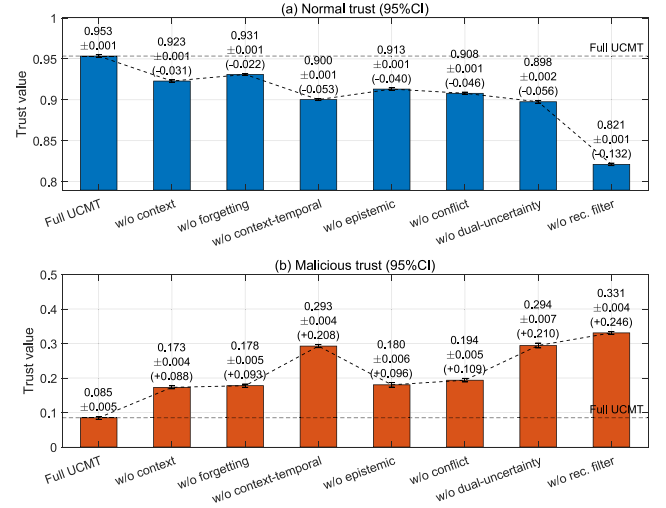


Fig. 14. Component ablation analysis.

5.6. Ablation analysis

An ablation study is conducted under the hybrid attack scenario by removing the core components of UCMT individually, with the comprehensive trust values of normal and malicious ESs used as the evaluation metric. The results in Fig. 14 show that removing any component degrades the evaluation performance. Specifically, after removing the forgetting factor and context similarity, the periodic camouflage behaviors of nodes are excessively memorized by historical records and cannot be reduced in time. The dual uncertainty fusion mechanism can reduce the weights of different trust sources according to evidence scarcity and source disagreement, thereby stabilizing the comprehensive trust. The recommendation filtering mechanism protects normal nodes. After its removal, normal ESs are contaminated by negative recommendations, resulting in decreased trust values. It is worth noting that removing individual factors, such as forgetting and epistemic uncertainty, has a limited impact on trust evaluation because these factors can compensate for each other. In contrast, removing them in groups leads to the most significant degradation. These results verify the core contributions of UCMT.

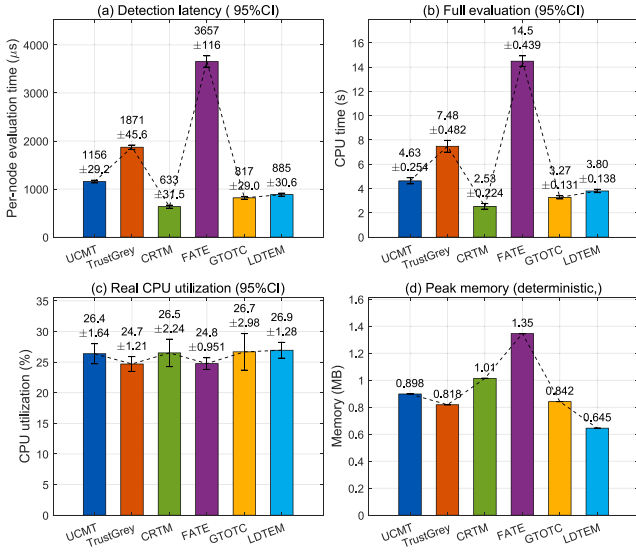


Fig. 15. Comparison of resource overheads.

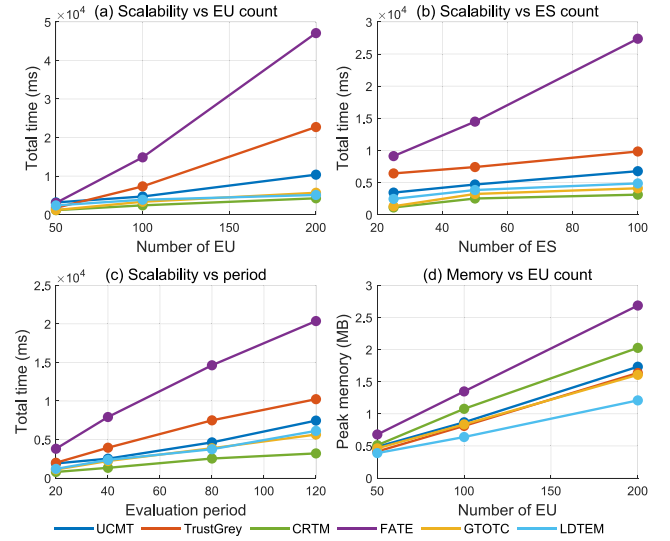


Fig. 16. Comparison of scalability.

5.7. Overhead analysis

To evaluate the practical deployment overhead of UCMT, its performance is measured against the baselines under the same environment in terms of detection latency, CPU time, CPU utilization, peak memory, and scalability with respect to network size and evaluation periods.

The results shown in Figs. 15 and 16 demonstrate that UCMT completes a single node trust evaluation at the millisecond level, while the CPU time of a single complete evaluation is approximately several seconds. Although UCMT performs multi-source evidence fusion and dual uncertainty modeling for each edge server in every evaluation period, these operations are all linear, and thus the overall overhead remains lightweight. In terms of CPU utilization, all methods exhibit similar moderate usage levels in the multi-core environment. Regarding memory consumption, the evaluation states maintained by different methods vary, but all remain small. In terms of scalability, the evaluation time of UCMT increases approximately linearly with the number of EUs, the number of ESs, and the evaluation periods. In contrast, FATE and TrustGrey exhibit quadratic complexity due to pairwise similarity iterations and PageRank power iterations, causing their evaluation time to increase rapidly with the number of EUs.

In summary, UCMT achieves scalability and deployability with near-linear computational and storage complexity, significantly lower overhead than quadratic-complexity methods. Meanwhile, it substantially improves trust evaluation accuracy and attack robustness.

5.8. Performance under offloading framework

To verify the effectiveness of UCMT in supporting collaborative operations in the CEE system, UCMT and the baseline models are deployed in the same computation offloading framework. This experiment investigates whether UCMT can transform its advantages in trust evaluation into system-level offloading benefits under the simultaneous presence of link noise, measurement noise, background load fluctuations, and malicious resource misreporting by ESs. The six metrics in Fig. 17 include the offloading failure rate, SLA violation rate, malicious ES selection ratio, system utility, false positive rate, and false negative rate. The environmental conditions are configured as follows: transmission delay jitter following a normal distribution, the link packet loss rate is 2%, completion time and resource measurements contain reporting errors of $\pm 5\%$, and ES available resources fluctuate within $\pm 20\%$ due to background load.

Let $\mathbb{N}_c$ and $\mathbb{N}_m$ denote the sets of users whose tasks are offloaded to the CSC and EU, respectively. p_m represents the unit computation resource cost of ES, p_0 represents the unit computation resource cost of CSC, $\tau_n - t_n$ denotes the total latency reduction, ω is the revenue conversion coefficient, $|\mathbb{B}_m|$ represents the number of sub-bands, and $h_{n,0}$ and $h_{n,m}$ denote the amounts of computation resources committed by the CSC and ES for task execution, respectively. The offloading optimization objective is formulated by minimizing the difference between the resource cost and the expected revenue, as follows:

$$\min \left(p_0 \sum_{u_n \in \mathbb{N}_c} h_{n,0} + \sum_{e_m \in E} p_m \sum_{u_n \in \mathbb{N}_m} h_{n,m} - \sum_{u_n \in \mathbb{N}_m} \omega (\tau_n - t_n) \right). \quad (24)$$

$$h_{n,0}, h_{n,m} > 0 \quad u_n \in U \quad (C1)$$

$$|\mathbb{N}_m| \leq |\mathbb{B}_m| \quad e_m \in E \quad (C2)$$

$$\sum_{u_n \in \mathbb{N}_m} h_n \leq f_m \quad e_m \in E \quad (C3)$$

$$t_n \leq \tau_n \quad u_n \in \mathbb{N}_m, e_m \in E \quad (C4)$$

The above optimization objective is subject to constraints C1–C4. Constraint C1 indicates that the computation resources allocated to each task are non-negative. Constraint C2 ensures that the number of simultaneously served users at each ES does not exceed its total available number of sub-bands, and that different EUs associated with the same ES are assigned mutually exclusive sub-bands. Constraint C3 indicates that the total computation resources allocated by an ES to all tasks cannot exceed its available idle computation resources. Constraint C4 represents the latency constraint, requiring that the actual completion time t_n of each task does not exceed its maximum tolerable latency τ_n .

In trust-based offloading decision-making, this paper follows the TCO framework (Wang et al., 2025a) and decomposes the original offloading problem into three stages: trust-based user association, trust-based feasibility checking and sub-bands allocation, and computation resource allocation based on optimization objective reduction. Since this paper mainly focuses on the trust model, the detailed decision-making process is referred to Wang et al. (2025a). In general, the trust evaluation results affect the decision-making process through resource requirements, latency feasibility, and candidate offloading benefits. At the offloading decision layer, the system employs a trust-weighted utility function to quantify the net benefit of migrating tasks from the CSC to ESs and allocates subchannels according to the descending order of ES trust values. At the resource allocation layer, the system calculates the actual allocated resources weighted by trust values and associates the minimum computation resource requirement with the corresponding trust values.

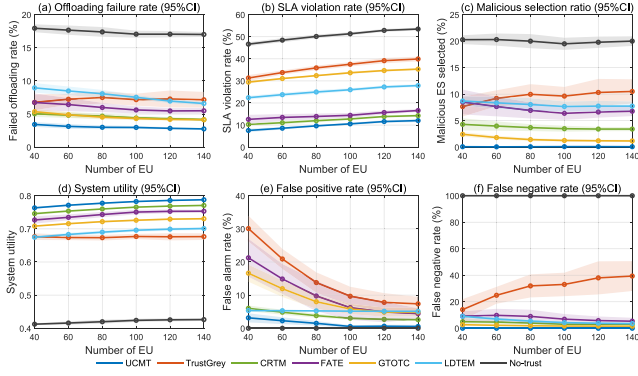


Fig. 17. Computation offloading under trust-based assistance.

All trust models use the same offloading framework, with the only difference being the method used to calculate ES trust values. The default trust threshold is set to 0.5 for all trust models, which is used for malicious node identification and the calculation of false positive and false negative rates.

TrustGrey has gray buffering and historical smoothing characteristics, resulting in slower responses to abrupt or persistent resource misreporting behaviors. Therefore, it exhibits higher malicious ES selection ratios and SLA violation rates. FATE focuses on reputation and feedback filtering, but it tends to lag behind when malicious reporting and resource advertisement disguises exist. CRTM has certain capability in handling collusion and recommendation credibility, leading to relatively better performance on some detection metrics; however, it lacks fine-grained direct constraints on resource misreporting and edge execution quality. GTOTC mainly focuses on coarse-grained success/failure modeling and cross-domain trust propagation, making it insensitive to actual completion-time degradation caused by SLA violations and resource misreporting. Therefore, although it can avoid some malicious nodes, it cannot achieve the highest system utility. LDTEM cannot resolve the separability issue between noise and malicious behaviors when noise and resource misreporting coexist. The trust-free approach does not suffer from false positives, but it is also completely unable to identify malicious ESs. UCMT does not simply reduce the trust of malicious nodes; instead, it feeds malicious resource misreporting and execution anomalies back into offloading decisions, thereby reducing the probability of assigning tasks to unreliable ESs. The results demonstrate that UCMT achieves superior system-level offloading benefits in the constructed realistic noisy simulation environment.

Since the UCMT model itself does not rely on a trust threshold, a threshold is introduced only in the offloading scenario to support offloading decisions. Therefore, a dedicated threshold sensitivity analysis is conducted for UCMT. The results are shown in Fig. 18.

Three levels of relative noise intensity are considered in this experiment, namely Low Noise, Nominal Noise, and High Noise, corresponding to scaling factors of 0.6, 1, and 1.6, respectively. The noise here represents a unified intensity scaling of multiple disturbances that affect trust evidence in the offloading environment. Specifically, the probability of occasional failures of normal ESs and EUs increases with the noise level, the upper and lower bounds of link rates deteriorate as the noise intensity increases, and the upper bound of temporal intervals also expands with increasing noise. In the figure, N denotes normal ESs, whereas M denotes malicious ESs.

UCMT outputs continuous comprehensive trust values, and 0.5 can be adopted as a conservative default threshold. However, in practical deployments, threshold calibration should be performed according to the recent trust distribution, environmental noise conditions, and task workloads. The results demonstrate that a clear separation between normal ESs and malicious ESs is maintained under low-noise and nominal-noise conditions. In contrast, high noise increases the trust

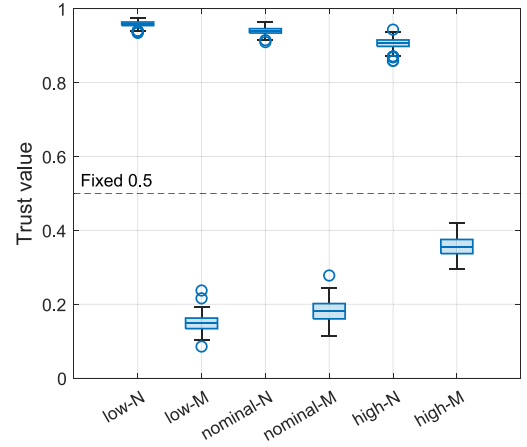


Fig. 18. Trust threshold analysis.

values of malicious ESs and raises the risk of false negatives when low thresholds are adopted.

5.9. Discussion

This subsection contains three parts: complexity analysis, adaptation to environmental dynamics, and limitations.

5.9.1. Complexity analysis

The computational complexity of UCMT consists of four main components: multi-source trust, recommendation analysis, trust fusion, and trust update. Let the number of EUs be N , the number of ESs be M , the number of EUs that have historical interactions with the target ES be $N_{U,E}$, the number of ESs that have collaborated with the target ES be N_c , the length of historical interaction records of an ES be L , the number of historical records of an EU be K , and the number of historical trust records for backtracking be J . Let E_{ES} be the number of edges in all ES-ES interaction graphs, and let E_{EU} be the number of edges in all EU-ES interaction graphs.

For CSC-side computation, evaluating resource sufficiency, consistency, and task timeout rate requires a time complexity of $O(N+L)$. For each host ES, computing response timeliness, task execution efficiency, and collaboration familiarity has a constant complexity of $O(1)$, and aggregating all host ES evaluations incurs a complexity of $O(N_c)$. For EU-to-ES trust computation, all EUs that have interacted with the target ES must be traversed, resulting in a complexity of $O(N_{U,E})$. In the trust fusion module, both uncertainty estimation and aggregation are performed in constant time. In the trust update stage, historical trust over J periods is aggregated with complexity $O(J)$. Since $J \ll N$, this term can be regarded as negligible in large-scale settings. The CSC-to-EU trust computation requires $O(K)$. Therefore, the total time complexity of computing the trust value for a single ES is $O(N + L + N_c + N_{U,E} + K)$. The total time complexity of the full-system is $O(\sum_{m=1}^M N + L_m + N_{c,m} + N_{U,E,m} + K_m)$, which can be further expressed as $O(M \cdot N + L_{total} + E_{ES} + E_{EU} + K_{total})$.

In the worst-case scenario (i.e., a fully connected graph), $E_{ES} = O(M^2)$ and $E_{EU} = O(M \cdot N)$. Therefore, the full-system complexity is approximately $O(M^2 + M \cdot N)$, exhibiting nonlinear growth. However, in practical CEE systems, due to constraints such as geographical distance, network latency, and service coverage, ES nodes only cooperate with nearby nodes, and EUs only connect to a limited number of ES nodes. Consequently, the interaction graph generally exhibits sparsity. Under sparse conditions, $E_{ES} \approx O(M)$ and $E_{EU} \approx O(N)$, reducing full-system complexity to an approximately linear scale.

The space complexity is mainly determined by the storage of historical interaction records, recommendation records, and contextual

information. These storage requirements scale with $\mathbb{N}_c$ and $\mathbb{N}_{U,E}$. Therefore, the total space complexity is simplified to $O(\mathbb{N}_c + \mathbb{N}_{U,E})$. In the worst case, all ES nodes are mutually connected, and each EU interacts with every ES node. Consequently, the total space complexity reaches $O(M^2 + M \cdot N)$. For typical CEE systems with sparse interaction patterns, the space complexity decreases to $O(M + N)$.

5.9.2. Adaptation to environmental dynamics

A key practical requirement of CEE systems is the ability to adapt to changing conditions, including task type transitions, node behavior variations, network condition fluctuations, and the evolution of evidence availability. UCMT addresses these changes through two mechanisms within the trust model.

At the trust fusion level, epistemic uncertainty and conflict uncertainty modulate the contribution of each trust source according to environmental conditions. Epistemic uncertainty reduces the reliance on a trust source when evidence is sparse, preventing trust decisions from being dominated by limited samples. Conflict uncertainty detects disagreements among the three trust sources and automatically reduces the weight of abnormal sources. This mechanism is effective in handling trust inconsistencies caused by network congestion or measurement noise that disrupts observations from specific sources.

At the trust update level, context similarity and the forgetting factor jointly control the influence of historical records on current trust evaluation. When the task context changes, context similarity automatically filters dissimilar historical records, ensuring that only historical experiences under similar conditions participate in the current trust update. Meanwhile, the forgetting factor introduces a recency bias, where trust values from older interactions decay faster than those from newer interactions. This property is crucial when nodes switch between benign and malicious behaviors. This dual mechanism enables UCMT to track the temporal evolution of behaviors without being excessively anchored to outdated observations.

5.9.3. Limitations

Although UCMT demonstrates superior performance under various attack scenarios, its limitations in terms of single-point centralization, data privacy, and generalization capability against unknown attacks should be acknowledged. In accordance with the trust anchor assumption presented in Section 3.1, the robustness and attack resistance of UCMT reported in the experimental results are evaluated assuming that the CSC faithfully fulfills its trust evaluation role. The following discussion on CSC-related limitations addresses the potential consequences in scenarios where this assumption is violated.

This work assumes that the CSC is a trusted central node, which serves as the logical foundation for establishing trust evaluation. However, this assumption may face challenges in practical deployments, with the following potential impacts. If the CSC is compromised, the objective scores in Eq. (5) and the recommendation weights in Eq. (12) can both be tampered with, resulting in distorted comprehensive trust values. If the CSC becomes unavailable, both Eq. (5) and the recommendation weights are unavailable, and the model degrades to relying only on ES-to-ES collaborative trust and EU direct trust. Although basic evaluation can still be maintained, the capability to defend against recommendation attacks is lost. If the CSC response is delayed, trust scores for the current period cannot be obtained in time, and the system may make offloading decisions based on outdated trust values during the delay period. If the CSC is partially controlled or contains compromised internal nodes, inconsistencies may occur in its trust outputs. Although the uncertainty fusion mechanism can mitigate some negative impacts caused by an unreliable CSC, its effectiveness is limited if the CSC itself participates in collusion.

From the perspective of data privacy, UCMT requires fine-grained interaction data, such as task completion time, resource allocation amount, and transmission rate, for trust evaluation. These data involve user privacy and are subject to regulatory constraints in cross-domain deployments. Currently, UCMT centrally stores raw interaction

logs at the CSC. Once the CSC is compromised, large-scale privacy leakage may occur, and the model lacks protection mechanisms such as differential privacy or secure multi-party computation. Furthermore, UCMT relies on predefined suspicious behavior patterns to defend against known attacks. This prior-knowledge-based defense logic has fundamental limitations when facing zero-day attacks or adaptive attacks.

The above limitations are inherent characteristics of current trust evaluation architectures and represent important directions for our future work. Moreover, it should be emphasized that the robustness evaluation presented in this paper is primarily based on simulation environments and has not been validated across diverse hardware architectures, network protocol stacks, or real-world edge deployment scenarios. Consequently, the direct transferability of the quantitative parameter configurations across different CEE systems remains an open research issue.

6. Conclusion

Due to their proximity to end users and their potential involvement in processing sensitive data, ESs must be trustworthy. In this paper, a multi-source trust model based on uncertainty and context-temporal awareness, namely UCMT, is proposed. It provides an objective evaluation from the perspectives of long-term stability, cooperative reliability, and service quality, enabling comprehensive trust evaluation of ESs. In addition, the proposed trust aggregation and update mechanisms effectively maintain positive interaction dynamics among nodes. Extensive experimental simulation results demonstrate that UCMT outperforms baseline models in terms of evaluation accuracy, robustness, and response speed. It also exhibits higher stability under various attack types and system scales. In future work, adaptive perception, intelligent anomalous behavior learning, and cross-domain distributed trust collaboration will be focused on. Meanwhile, decentralized multi-controller consensus schemes will also be explored to avoid reliance on a single CSC, thereby further improving the robustness and practicality of the model in complex CEE environments.

CRedit authorship contribution statement

Minglong Cheng: Writing – original draft, Methodology, Conceptualization. **Wei Chen:** Investigation, Funding acquisition. **Chen Huang:** Visualization, Formal analysis. **Weidong Fang:** Methodology, Investigation. **Tingting Xu:** Writing – review & editing, Software. **Ziyang Xing:** Visualization, Software. **Zehua Wang:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary materials for research data and experimental setup. Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cose.2026.105163>.

Data availability

The authors do not have permission to share data.

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